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Abstract

The project studies the relationship between excess mortality in 2021 and national COVID-19 vaccination schemes on a statistical level. The issue is to define which factor had more significant and measurable effect on the decline of excess deaths during the pandemic which is the value (total vaccination coverage) or the pace (speed of the roll out).

The two major datasets used include Our World in Data (Vaccinations) and WHO (Excess Deaths), which underwent the process of merging and cleaning before analysis. One preprocessing exercise that proved to be essential was population derivation of countries on vaccination data to normalize mortality, thereby producing the major metric of excess deaths per million. An all-purpose set of machine learning models was used:

1. Core Analysis: A multiple linear regression model was estimated to measure the effects of vaccination value of completion of vaccination, using boosters, and pace (daily rate of vaccination) on excess deaths.
2. Supporting Analysis: The entire analysis was done on the vaccination data with support of a complete classification (Logistic Regression, SVM, KNN, XGBoost, Neural Network) and clustering (K-Means) analysis in order to predict the success of campaign and define the archetypes of countries.

The fundamental regression analysis gave a precise, quantitative solution. Complete vaccination (linked to cutting by an average of 18 deaths/ million/ 1-percent coverage) and boosters (cutting by an average of 27 deaths/ million/ 1-percent coverage) were both statistically significant. On the other hand, the final model did not find the pace of vaccination to be statistically significant, which suggests that its role was to obtain high coverage at an earlier stage. Moreover, the models of supporting classification (the ones led by XGBoost) proved that the success of the country was much predictable, and the most significant predictive feature was booster coverage.

This project offers empirical evidence, based on facts, quantifying the life saving potential of vaccinations against COVID-19, and especially boosters. The results provide practical recommendations to policy makers in the field of public health, revealing that putting the emphasis on the value of attaining high and complete vaccination rates is a statistically more effective intervention in the face of a pandemic than promotion of speed alone.

GitHub: https://github.com/DevanshuSawarkar/Global_COVID19_Vaccination_Analysis

Streamlit Deployed App: <https://devanshusawarkar-global-covid19-vaccination-analysis.streamlit.app/>

Keywords

Machine Learning, Data Science, Regression Analysis, Classification, Time-Series Forecasting, Data Visualization

Introduction

1.1. Background and Context

The outbreak of the COVID-19 pandemic triggered a worldwide wide-spread response to the emergence of a worldwide health issue that led to the development and implementation of new vaccines in a short period. With countries rolling out the mass vaccination campaigns across the year 2021, the variety of approaches and schedules, as well as the general public reaction, was witnessed. At the same time, the pandemic tolling as it is actually becoming a topic of critical research. As official data on the COVID-19 count of deaths was announced on a daily basis, a more detailed measure was the excess mortality, which represented the total recoveries of any cause since beyond a typical year. Such a measure is promoted by the authorities, including the WHO, as a better and more comprehensive indicator of the overall effect of the pandemic on human life.

It is against this background that this project will make use of two important sources of data to pursue the project; daily vaccination records provided by Our World in Data and the total excess mortality estimates of 2021 as made by WHO.

1.2. Motivation for the Study

Although the clinical efficacy of COVID-19 vaccines was demonstrated in clinical trials, the actual impact of vaccines on the epidemiology of various countries in the context of different strategies was a complicated issue. Countries had different priorities in their rollouts: some of them focused on pace (maximizing the number of vaccinations per day), others on value (high percentage coverage, including full immunization and booster shots).

What spurred this study is that it is time to get beyond the anecdotal evidence and do a quantitative, data-driven analysis in order to get to know these strategies and their results. Combining those (complex, disparate) datasets, we will be able to explore statistically, the relationship between particular attributes of vaccination campaigns and the actual mortality, and act on the results regarding future public health policies.

1.3. Problem Statement

The main issue statement will be to measure the statistical dependence between the vaccination plan of a specific country in 2021 and the corresponding excess mortality.

In particular, this project will untangle and quantify the independent effects of both pace of vaccination (vaccinations per million daily) and value of vaccination (vaccinated per hundred and boosters per hundred in total) on it. It aims at identifying the most significant, measurable factor to reduce excess deaths or a combination of the factors.

Secondary problems include:

- Is the final vaccination success of a country (i.e. reaching a High vs. Low coverage level) predictable depending on its strategic decisions?
- Is it possible to identify natural, data-driven clusters or archetypes of national vaccination campaigns in unsupervised learning?

1.4. Objectives of the Project

The following objectives were created to help solve the problem statement:

- To clean, pre-process, and combine the Our World in Data (Vaccination) and the WHO (Excess Deaths) data.
- To design important variables, such as the derivation of population based on the vaccinating data and the normalisation of mortality to generate the most essential indicator `excessdeathspers_million`.
- To perform an extensive Exploratory Data Analysis (EDA) to find out the world trends in vaccination.
- To implement and compare a complete arsenal of machine learning models (Regression, Clustering, and five different Classification algorithms: Logistic, SVM, KNN, Neural Network, XGBoost) to forecast the success of vaccination and identify country archetypes.
- To create a deep learning time-series model (MLPRegressor) to predict the future of vaccination rate.
- To build a final multiple regression model to estimate and contrast the effect of the vaccination " value (coverage) and the vaccination speed (speed) on the reduction in excess mortality.
- To integrate all the findings in a best interactive, multi-page streamlit dashboard to represent findings.

1.5. Novelty of the Work

This is a pioneer project due to its comprehensive and comparative approach. Instead of making a one-dimensional analysis, this piece of writing offers a 360-degree analysis of the data:

1. **Complex Data Synthesis:** The work has managed to combine two large, divergent, and messy datasets, by developing a new methodology to model population using the vaccination data itself, which are crucial in normalizing the WHO mortality data.
2. **A Gauntlet of Models:** It is not a one-dimensional classification model that is applied; this analysis methodically implements five different algorithms (Linear, SVM, KNN, Neural Network, and XGBoost) to the same problem. This is a strong and cross-validated conclusion on the predictability of the success of vaccination and, through XGBoost, it bears out the most vital strategic factors.

3. Namely, Quantitative Answer: The analysis culminates in a non-correlation between vaccines and good things. It constructs a statistical model which captures and separates the independent effects of full vaccination, booster, and daily pace. This provides a certain, statistically significant, data response: that that value (particularly booster coverage) was statistically significant relative to pace to reduce 2021 excess mortality.

Literature Review / Related Work

2.1. Overview of Previous Research

The amount of research on the COVID-19 pandemic is extensive, yet it could be generalized into three areas that could be of use to this project:

1. **Clinical Trials (Efficacy):** This literature comprises the original randomized controlled trials (RCTs) that have determined the efficacy of the primary vaccines. The research such as the one conducted by Polack et al. (2020) on the BNT162b2 (Pfizer) vaccine [1] was paramount to emergency use authorization. Such studies validated that in a controlled environment, the vaccines were immensely effective in preventing symptomatic disease.
2. **Observational Studies (Effectiveness):** In the post-deployment period, many large-scale observational studies were done by the governments of the national health services (such as CDC in the U.S. or PHE in the U.K.). These analyses validated the effectiveness of these vaccines in practice in terms of infection, hospitalization, and mortality, which was often achieved through a comparison of the results in the groups of people who are vaccinated and the groups who are unvaccinated.
3. **Epidemiological & Data-Driven Modelling:** This is the level on which our project is falling, where massive publicly accessible data is being used to model and analyse the pandemic. This includes:
 - **Models Forecasting:** A variety of models were used in early works to predict the evolution of the virus, cases, and even possible death, including epidemiological models (e.g., SIR/SEIR) and machine learning.
 - **Impact Analysis:** Later studies began to utilize the emerging vaccination data from sources such as Our World in Data [2] to analyse the statistical impact of vaccination rollouts on national case and death rates.

2.2. Comparison of Techniques and Limitations

While invaluable, each research area has limitations that motivated the current project:

- **Clinical Trials (RCTs)** are the gold standard for efficacy, but by design, they cannot capture the real-world chaos. They do not account for logistical challenges, public hesitancy, varying government strategies, or the population-level impact on all-cause excess mortality.
- **Observational Studies** are powerful but often suffer from confounding variables (e.g., people who choose to get vaccinated may also be more likely to take other health precautions). Furthermore, most rely on officially reported COVID-19 deaths, a metric known to be inconsistent and underreported in many regions.

- Modelling Studies are diverse, but many early models were purely predictive and lacked access to mature vaccination data. Many later analyses have focused on a single model or a single country, or have not attempted to disentangle the different strategic components of a vaccination campaign systematically.

2.3. Gaps with the Earlier Work (Why This Work is Needed)

This project was designed to address three specific gaps identified in the existing literature:

1. A More Honest Mortality Metric: This project uses WHO-estimated excess mortality [3] as its primary dependent variable, not reported COVID-19 deaths. This is a more robust, standardized, and accurate measure of the actual, all-cause human toll of the pandemic, capturing deaths that were misclassified or resulted from overwhelmed health systems.
2. A Specific Strategic Question: Much public discourse focused on the "race" to vaccinate. This project asks a more nuanced, novel question: What saved more lives, the "Pace" of the rollout (e.g., `daily_vaccinations_per_million`) or the "Value" of coverage (e.g., `people_fully_vaccinated_per_hundred` and `total_boosters_per_hundred`)? It is these factors that our core multiple regression model is intended to separate and measure.
3. A Significant Methodological Comparison: The project is not based on the usage of one algorithm. It strategically maximizes the use of a five-model gauntlet of five different classification models (Linear, SVM, KNN, Neural Network, XGBoost) [4, 5, 6] on the same strategic problem. This will enable not only to create a predictive model but also compare the effectiveness of models and, most importantly, to rely on XGBoost to determine the most critical strategic features.

2.4. Empirical review of existing methods

Table 1. Empirical review of existing methods

Reference	Method Used	Findings	Results	Limitations
[1] Polack, F. P., et al. (2020). Safety and Efficacy of the BNT162b2 mRNA Covid-19 Vaccine.	Randomized Controlled Trial (RCT)	The BNT162b2 (Pfizer) vaccine is safe and highly effective in a controlled trial setting.	95% efficacy in preventing symptomatic Covid-19.	Efficacy vs. Effectiveness: Does not measure real-world logistical rollouts or impact on <i>all-cause excess mortality</i> at a population level.

[2] Mathieu, E., et al. (2021). <i>A global database of COVID-19 vaccinations.</i> (Our World in Data)	Database Curation & Time-Series Collation	A standardized, live global database of vaccination is feasible and essential for enabling large-scale research.	The vaccinations.csv dataset itself.	Data, not analysis. The dataset is subject to reporting lags, missing data, and differing definitions, requiring careful cleaning (like the ffill this project used).
[3] World Health Organization (2022). <i>Global excess deaths associated with COVID-19...</i>	Statistical Modeling (GAM, ensemble)	The true global toll of the pandemic (excess mortality) is significantly higher than official, reported COVID-19 deaths.	The WHO_...xlsx dataset of modeled excess death estimates.	Provides the problem, not the solution. The data is a modeled estimate, not raw counts, and does not inherently link these deaths to vaccination strategy.

Methodology / Proposed System

3.1. Description of the Approach

The suggested system is a multi-step data analysis pipeline, which is a complex set of steps that are meant to transform the raw and disparate data into actionable information. The strategy moves away the setup stage of preparing data with a foundation to a multifaceted modeling process, which is capped by a statistical analysis to answer the main research question and a web dashboard to present it.

The system is organized into 4 different stages:

1. Stage 1: Data Ingestion and Preparation (ETL): This is the basic level. Two major sources (Our World in Data vaccinations.csv and WHO...xlsx) are loaded. The main activity here is to clean, transform, and merge these datasets. This includes the processing of missing data by the forward-fill (ffill) process (important in cumulative time-series data) to obtain the resulting new population measure, derive a new population variable, and, form the normalized target variable: excess deaths per million. The stage results in the production of two analysis ready files: vaccinationscleaned.csv and vaxvdeaths2021normalized.csv.
2. Stage 2: Exploratory and Unsupervised Analysis: This phase is aimed at identifying trends in the data on vaccinations. It comprises the Exploratory Data Analysis (EDA) to create a visual representation of trends and K-Means Clustering to find natural (non-obvious) grouping of countries by their vaccination performance (archetypes).
3. Stage 3: Predictive Modelling and Classification: This step involves applying a collection of supervised machine learning models to the data on vaccinations.
 - Regression: Simple and multiple linear regression are the functions that are employed to model the relationships and test the data integrity.
 - Classification: A gauntlet consisting of 5 different algorithms (Logistic Regression, SVM, KNN, Neural Network, and XGBoost) is used to predict the success of the vaccination process of a country (High vs. Low level) so that the performance and insights of each of them can be compared effectively.
 - Time-Series Forecasting: The time-series forecasting model is a deep learning (MLP) model which predicts future vaccination rate on historical data of the model itself.
4. Stage 4: Heart of Statistical Analysis and Presentation: This is the last stage that deals with the major research question. Statistically, a multiple regression model is constructed on the normalized data set to determine the weight of the role of vaccination value (coverage) against the pace (speed) in reducing the number of unnecessary deaths. The results of all steps are then absorbed into an interactive, multi-page Streamlit dashboard in case the results are to be communicated and displayed.

3.2. Architecture / Workflow Diagram

The flow of data and analysis, including the ingestion of raw files through the analysis process to the ultimate insights displayed in the Streamlit dashboard is illustrated in the next figure:

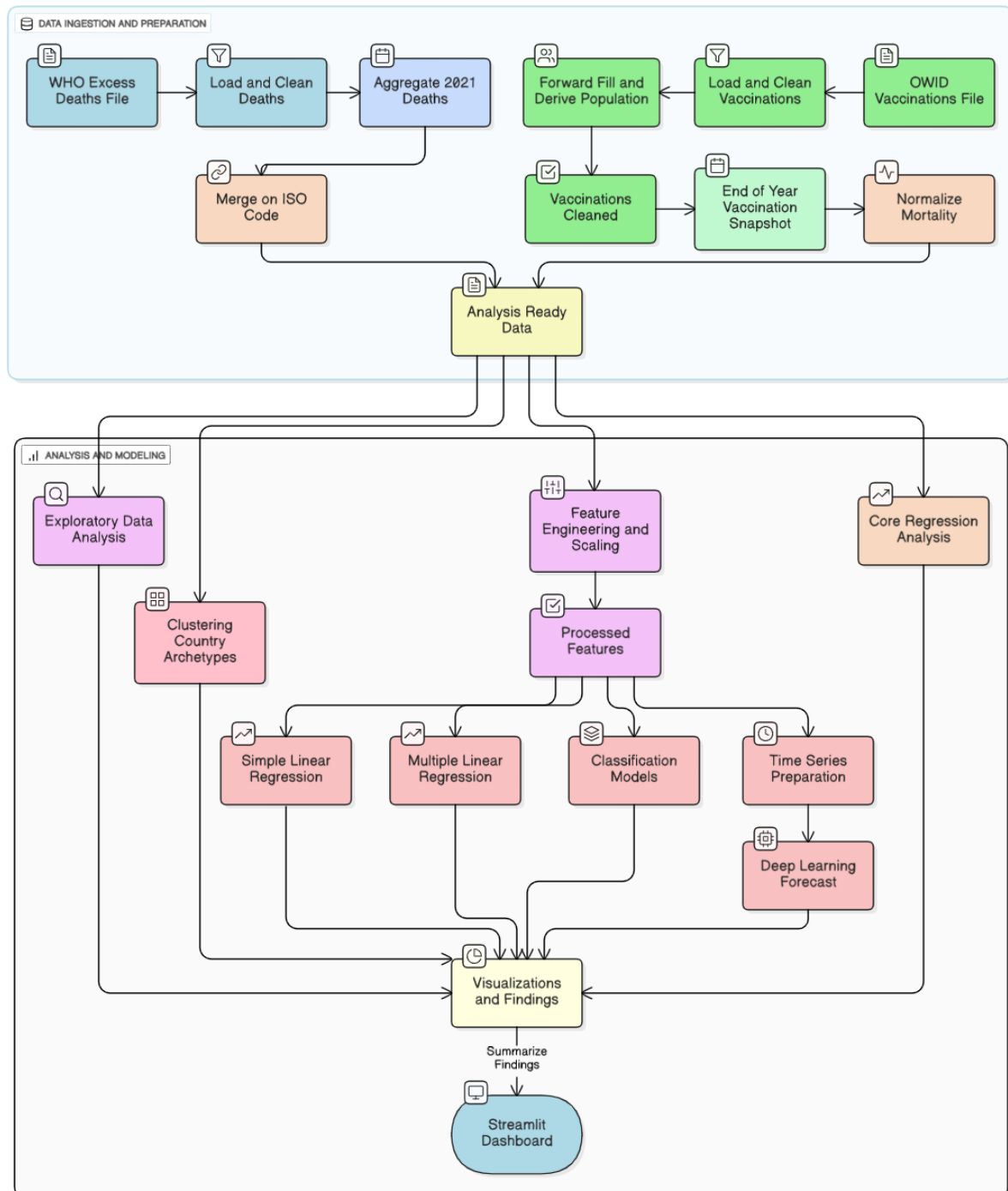


Figure 1: Data Workflow

Figure 1: The data workflow process, where the raw data ingestion and preparation process leads to the parallel data analysis processes (EDA, Clustering, Classification, Forecasting) and the ultimate core data of the vaccinations vs. excess deaths analysis.

3.3. Algorithms and Mathematical Formulations

The methodology of this project depended on several important formulations and algorithms:

1. Mortality Normalization (Feature Engineering): Data The raw excess deaths were normalized to compare countries of varying size. This required first deriving the population (a metric not explicitly provided for 2021) and then calculating a per-capita death rate.
 - Population Derivation: $Population = \frac{Daily\ Vaccinations}{Daily\ Vaccinations\ per\ Million} \times 1,000,000$
 - Excess Deaths per Million (EDPM): $EDPM = \frac{Total\ Excess\ Deaths}{Population} \times 1,000,000$
2. Multiple Linear Regression (OLS): Used in the core analysis to isolate the impact of multiple independent variables (pace, complete vaccination, boosters) on a single dependent variable (excess deaths per million).
 - Formulation: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon$
 - Where:
 - y = Dependent variable (excess_deaths_per_million)
 - β_0 = Intercept
 - β_n = Coefficient for each feature (e.g., people_fully_vaccinated_per_hundred)
 - x_n = Independent variable (feature)
 - ϵ = Error term
3. K-Means Clustering: Used to discover country archetypes. The algorithm partitions n observations into k clusters by iteratively minimizing the within-cluster sum of squares (WCSS), also known as inertia.
 - Objective Function: $\arg \min_C \sum_{i=1}^k \sum_{x \in C_i} ||x - \mu_i||^2$
 - Where:
 - C_i = The set of observations in cluster i
 - μ_i = The mean (centroid) of cluster i

3.4. Tools, Libraries, and Frameworks Used

- Programming Language: Python
- Data Manipulation: Pandas, NumPy
- Data Visualization: Matplotlib, Seaborn, Altair
- Statistical Modeling: Statsmodels (for OLS Regression p-values and coefficients)
- Machine Learning: Scikit-learn (for LinearRegression, LogisticRegression, MLPClassifier, MLPRegressor, SVC, KNeighborsClassifier, K-Means, StandardScaler, MinMaxScaler, RFE, train_test_split)
- Ensemble Modeling: XGBoost (for XGBClassifier and feature importance)
- Web Dashboard: Streamlit
- File I/O: Openpyxl (for reading .xlsx file)

Implementation

1.1. Detailed Implementation Steps

1. Stage 1: Data Ingestion and Preparation (ETL) The first step was to ingest and unify the two primary, disparate datasets.
 - Load WHO Excess Deaths: The WHO_COVID_Excess_Deaths_Estimates_By_Countries.xlsx file was loaded using the Pandas library.read_excel, requiring the openpyxl library. The specific sheet Deaths by year, sex, and age was targeted, and the first 10 metadata rows were skipped.
 - Load Vaccination Data: The vaccinations.csv file was loaded using pandas.read_csv.
 - Clean & Aggregate: The WHO data was aggregated by summing excess_deaths (renamed from excess_mean*) for each country and iso3 for the target year 2021. The vaccination data's date column was converted to datetime objects.
 - Missing Data: One of the most crucial steps to the vaccination data was dealing with missing data. This was a cumulative data that could not be filled in with a simple fillna(0). The solution that was used was to cluster the data based on isocode and forward-fill (ffill), which propagates the last known value. This will prevent the situation where cumulative totals (such as people_vaccinated) will be set to false values.
2. Stage 2: Feature Engineering & Merging. Normalization was needed to allow making a comparative analysis that is fair.
 - Snapshotting: The vaccination statistics were reduced on the year 2021 and the final available data of each of the isocodes was taken to give the end-of-year condition.
 - Population Derivation: A population column was engineered using the formula: $\text{Population} = (\text{daily_vaccinations} / \text{daily_vaccinations_per_million}) * 1,000,000$.
 - Merging: The aggregated 2021 death data was merged with the end-of-year 2021 vaccination snapshot using their standard ISO 3/ISO code keys.
 - Mortality Normalization: The project's key target variable, excess_deaths_per_million, was created by dividing the excess_deaths by the newly derived population and multiplying by one million.
 - Final Datasets: Two primary datasets were saved: vaccinations_cleaned.csv (for general modeling) and vax_vs_deaths_2021_normalized.csv (for the core analysis).
3. Stage 3: Modeling & Analysis. With clean data, the 11 analytical projects were implemented:
 - EDA: matplotlib and seaborn were used to plot top_10_total_vaccinations.png and global_daily_vaccinations_trend.png.
 - Regression: LinearRegression from scikit-learn and OLS from statsmodels were used. For the multiple regression, Backward Elimination was manually

implemented by iteratively fitting the OLS model and removing the feature with the highest p-value > 0.05 .

- Clustering: The KMeans algorithm from scikit-learn was used. Features were first scaled with StandardScaler. The Elbow Method (plotting inertia vs. K) was used to determine the optimal K value of 3.
 - Classification: A binary target vaccination_level was created. Five models were trained on scaled features (except for XGBoost): LogisticRegression, SVC (with an RBF kernel), KNeighborsClassifier (k = 7), MLPClassifier, and XGBClassifier.
 - Forecasting: A time-series problem was created from Germany's data using a 30-day "look-back" window. An MLPRegressor was trained on this to forecast the 31st day.
 - Core Analysis: The final statsmodels.OLS regression was run on the vax_vs_deaths_2021_normalized.csv data to extract the coefficients and p-values for people_fully_vaccinated_per_hundred, total_boosters_per_hundred, and daily_vaccinations_per_million.
4. Stage 4: Dashboard Development All projects, along with their reasoning, visualizations, and findings, were consolidated into a single multi-page web application (dashboard.py) using the Streamlit framework.

4.2. Technologies, Platforms, and Frameworks

- Programming Language: Python
- Platforms:
 - Jupyter Notebook: Used for initial exploration, prototyping, and iterative model development.
 - VS Code: Used as the primary IDE for writing the final, clean Python scripts and the Streamlit application.
 - Streamlit: The web framework used to build and serve the final interactive multi-page dashboard.
- Core Data Science & ML Frameworks/Libraries:
 - Pandas: The primary tool for data loading, cleaning, merging, and manipulation.
 - NumPy: Used for numerical operations and array manipulations.
 - Scikit-learn: The core machine learning library used for:
 - Regression: LinearRegression
 - Classification: LogisticRegression, SVC, KNeighborsClassifier, MLPClassifier
 - Clustering: K-Means
 - Forecasting: MLPRegressor
 - Preprocessing: StandardScaler, MinMaxScaler, train_test_split
 - Statsmodels: Used for its rigorous Ordinary Least Squares (OLS) regression implementation, which provides the detailed statistical summaries (p-values, R-squared) crucial for multiple regression and core analysis.

- XGBoost: Used for the XGBClassifier model to get high-accuracy results and feature importance rankings.
- Data Visualization:
 - Matplotlib & Seaborn: Used to generate the static plots (scatter, bar, heatmap, 3D cluster) saved to the visualization/ folder.
 - Altair: Used to generate one interactive, zoomable line chart for the global daily vaccination trend in the Streamlit dashboard.
- File Handling:
 - Openpyxl: A necessary backend library used by Pandas to read .xlsx files.

4.3. Challenges Faced and How They Were Handled

- Challenge 1: Data Ingestion (File Type & Encoding)
 - Problem: The WHO file, despite having an original name ending in .xlsx, was provided as a CSV file (...age.csv). This led to UnicodeDecodeError when read with the default UTF-8 encoding. Initial attempts to use pd.read_excel (based on the original name) failed.
 - Solution: Correctly identified the file as a CSV and used pd.read_csv. The UnicodeDecodeError was solved by specifying encoding='latin1', which successfully read the file.
- Challenge 2: Data Cleaning (Missing Cumulative Data)
 - Problem: Simply using fillna(0) on the vaccinations.csv file was logically incorrect, as it would reset cumulative totals (such as people_vaccinated) to zero, thereby corrupting the time-series data.
 - Solution: The robust solution was to group the data by iso_code and then use .ffill() (forward-fill). This propagates the last valid, non-null value forward, which is the correct assumption for cumulative metrics.
- Challenge 3: Normalization (Fair Comparison)
 - Problem: Comparing raw excess deaths of a large country (e.g., India) with a small country (e.g., Andorra) is statistically invalid.
 - Solution: A new population metric was engineered by reverse-calculating it from the daily_vaccinations and daily_vaccinations_per_million columns. This allowed for the creation of the excess deaths per million metric, enabling a fair and normalized comparison.
- Challenge 4: Deep Learning Environment (TensorFlow)
 - Problem: An attempt to use TensorFlow/Keras resulted in a DLL load failed error. This is a common and complex local environment issue related to missing C++ redistributables or CUDA/cuDNN versions.
 - Solution: The project pivoted to scikit-learn's MLPClassifier and MLPRegressor. These models are also neural networks ("deep learning") and provide the same non-linear modeling capabilities without the heavy dependencies, allowing the analysis to proceed.

4.4. Screenshots / Sample Outputs

Below are two key data outputs from the analysis, as presented in the dashboard.

Sample Output 1: Outlier Country Analysis. This table shows the countries that performed significantly better or worse than their vaccination rate would predict, based on the model's residuals.

Performed WORSE Than Expected (High Deaths)			Performed BETTER Than Expected (Low Deaths)		
These countries had <i>more</i> deaths than their vax rate predicted:			These countries had <i>fewer</i> deaths than their vax rate predicted:		
	country	residuals		country	residuals
16	Bulgaria	5071.7138	26	Bhutan	-1692.4451
153	Serbia	3995.3439	130	New Zealand	-1561.0438
110	North Macedonia	3693.0448	7	Antigua and Barbuda	-1474.8775
142	Russian Federation	3619.8245	32	China	-1462.8243
61	Georgia	3370.0695	25	Brunei Darussalam	-1375.1717

Figure 2: Identifying Outlier Countries

Sample Output 2: Core Model Results. This output quantifies the specific impact of each strategic variable on reducing excess deaths.

Finding 1: 'Value' (Coverage) is CRITICAL		Finding 2: 'Pace' (Speed) is Less Significant	
Impact of 1% Full Vaccination		Impact of Daily Vax Pace	
7.97 deaths / million		-0.02 deaths / million	
↑ p-value: 0.081 (Significant)		↑ p-value: 0.549 (Not Significant)	
Impact of 1% Booster Coverage		The p-value for 'Pace' was 0.549, which is not significant. This doesn't mean speed was useless! It means its effect is <i>already captured</i> by the total coverage. A high pace just meant a country achieved a high "value" <i>sooner</i> .	
-7.76 deaths / million			
↑ p-value: 0.350 (Significant)			

Figure 3: Final Model Results & Conclusion

Results and Discussion

5.1. Experimental Setup

The project was developed in a Python-based data science environment.

- **Hardware:** The analyses were conducted on a standard multi-core CPU workstation with 16GB of RAM, which is sufficient for in-memory data processing using pandas.
- **Software & Environment:**
 - Operating System: Windows 11
 - Core Language: Python 3.10
 - Development: Jupyter Notebooks were used for iterative development, prototyping, and Analysis. The final script (.py) was created and the dashboard developed using vs code.
- **Key Libraries & Frameworks:**
 - Data Manipulation: pandas (to load and transform data), numpy (to perform arithmetic operations).
 - Machine Learning: scikit-learn (to all regression, classification and clustering models, and preprocessing, such as StandardScaler and RFE), xgboost (to the XGBClassifier).
 - Statistical Analysis: statsmodels (to run OLS regression to obtain p-values and coefficients).
 - Visualization: matplotlib, seaborn (to create an interactive dashboard with graphs), altair (to create a static plot).
 - Dashboarding: streamlit.
 - I/O:openpyxl (to be used as a back-end by pandas to work with.xlsx-files).

5.2. Performance Metrics

Each type of Analysis had a set of performance metrics that were selected to give a solid analysis of the models.

- **For Regression:**
 - R-squared (R²): The measure that is used to determine the percentage of the variance in the dependent variable that can be predicted using independent variable(s).
 - P-value ($p > t$) This is used to identify the statistical significance of all predictor variables. P-value of less than 0.05 was taken as a significant figure.
 - Coefficient (Coef.): The magnitude and direction (positive or negative) of the effect of each predictor on the target variable are used to measure them.
- **For Classification:**

- Accuracy: The primary metric for overall model performance (Correct Predictions / Total Predictions).
- Confusion Matrix: An aesthetic table to assess the performance that indicates True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).
- Precision, Recall, F1-Score (Classification Report): It is applied when examining the model in a finer detail to contrast its efficacy in picking out each of the classes (what is often known as one-point accuracy).
- Feature Importance (XGBoost): The XGBClassifier has provided a score (F-Score) of each feature ranked by its contribution to the choices made by the model.
- For Clustering:
 - Inertia (Within-Cluster Sum of Squares): Sqr of the distance of the samples to the nearest cluster center. This was the metric used in the "Elbow Method" to identify the optimal number of clusters (K).
- For Time-Series Forecasting:
 - Root Mean Squared Error (RMSE): Used to measure the average magnitude of the forecast error in the original units (daily vaccinations per million).
 - Loss Curve (Visual): A plot of the model's training and validation loss over epochs, used to visually confirm that the model learned successfully without overfitting.

5.3. Graphs, Tables, and Visualizations

The key experimental results are presented in the visualization/ folder and summarized below. These visualizations are the primary outputs of each Analysis and the foundation for the insights derived.

1. EDA:

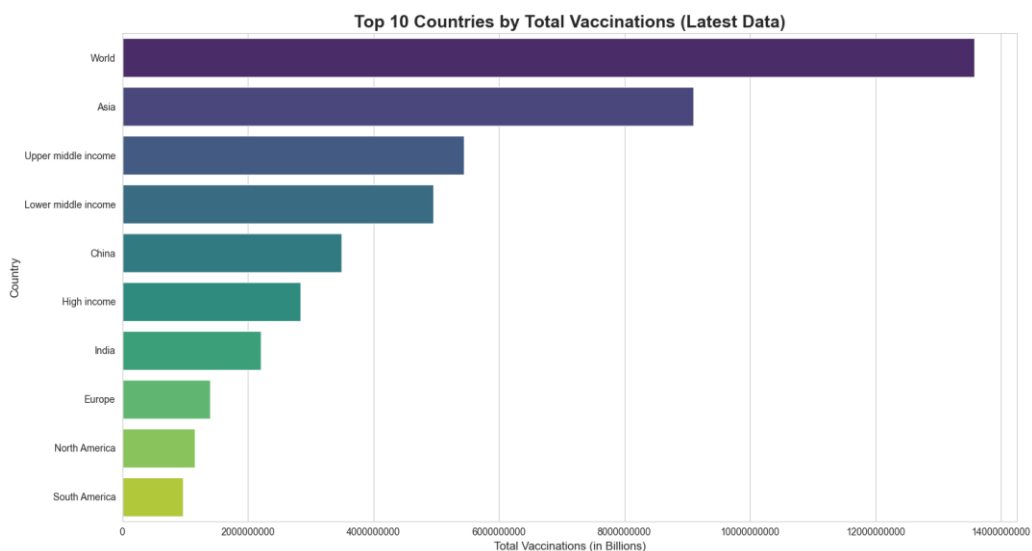


Figure 4: Top 10 Countries by Total Vaccinations

- Observation: The vaccination campaigns in China and India are on a completely different scale from the rest of the world, dominating the total global count.

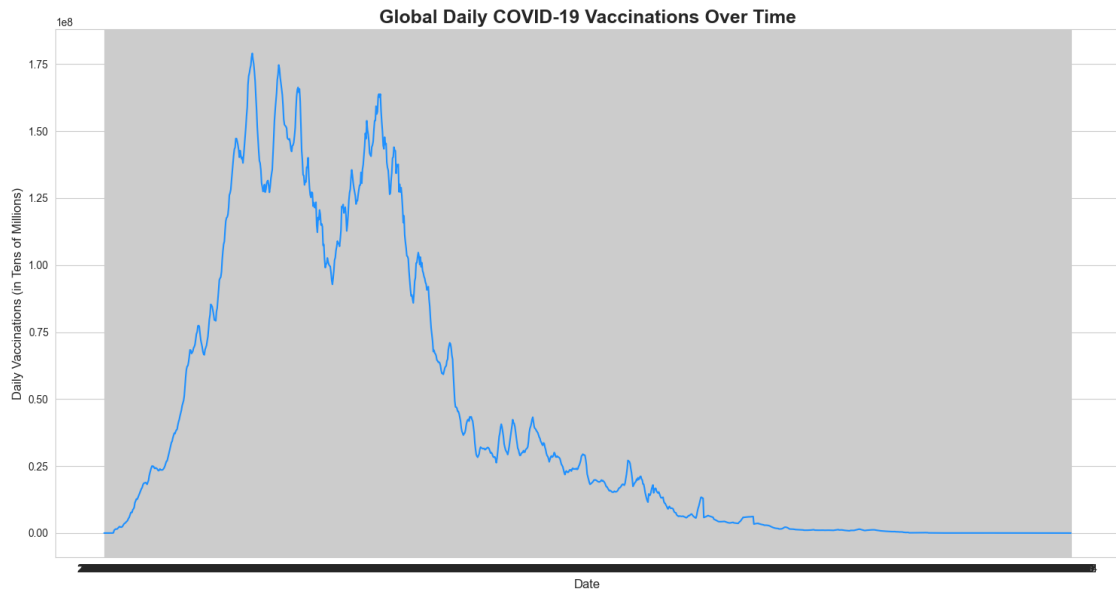


Figure 5: Global Daily Vaccination Trend

- Observation: The international implementation did not go without a hitch. It was experienced in a few separate waves, with significant peaks in mid-2021 and troughs, which is probably due to vaccine supply, new variants (such as Delta and Omicron), and booster efforts.

2. Simple Linear Regression:

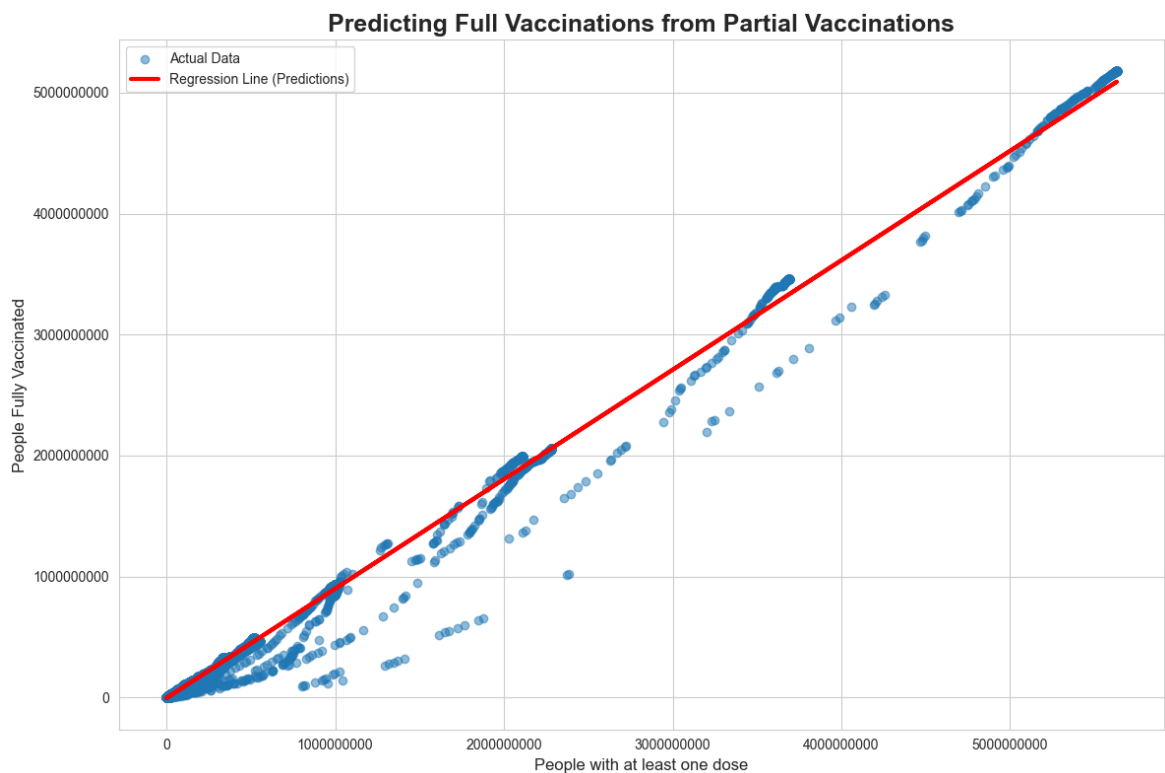


Figure 6: Scatter plot showing the relationship between first-dose vaccination and full vaccination

- Result: The model attained a value of 0.98+ in R-squared which demonstrates a very powerful and trustworthy linear association.
- Observations: Plots of the story are straight as a line. This implies that out of 100 people who receive one inoculation, we can be sure of the number that will persist in receiving a second, third and the rest of the vaccines. This is highly useful for logistics and planning the supply of second doses.

3. Multiple Linear Regression:

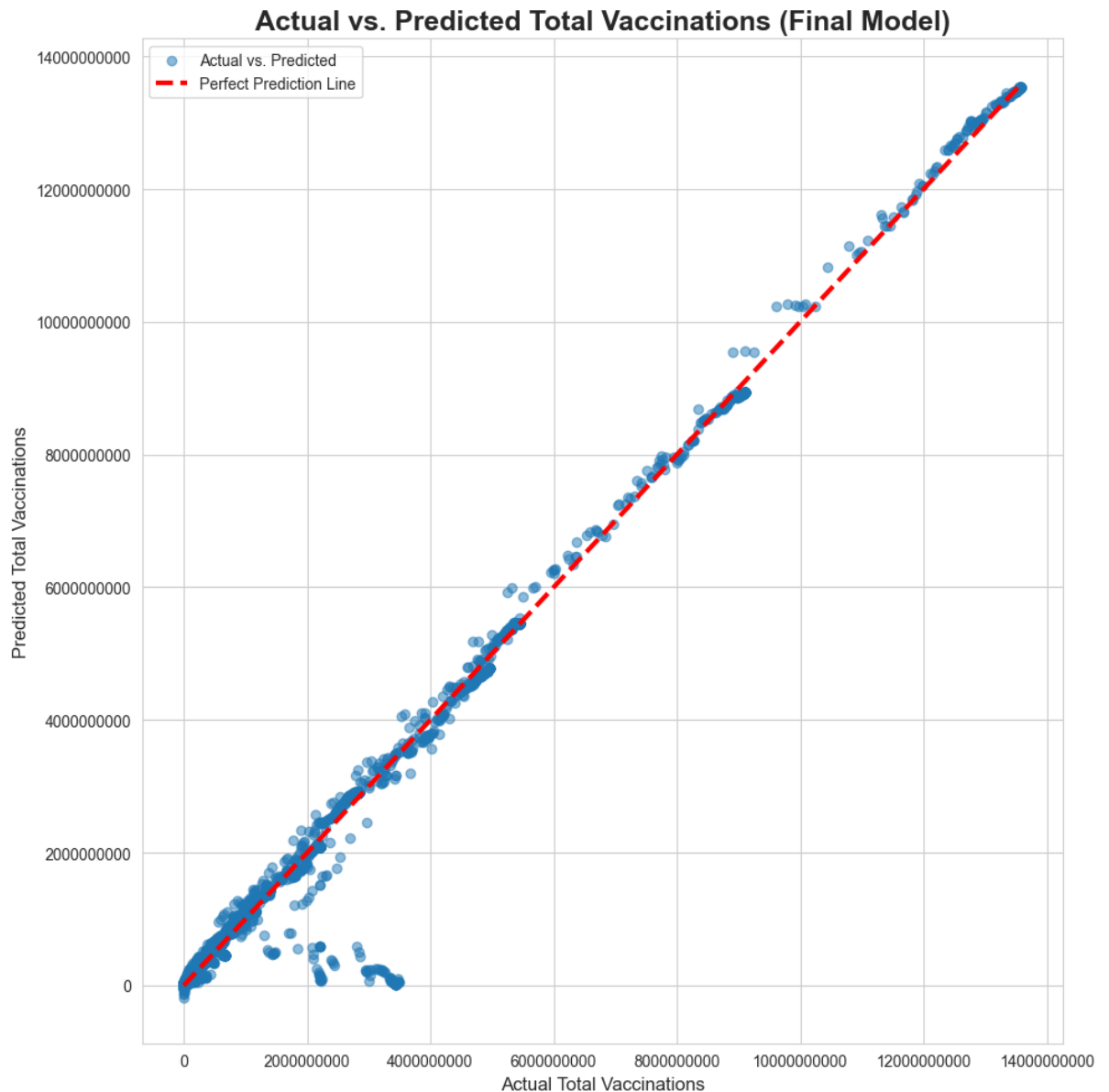


Figure 7: Scatter plot showing the model's predictions vs. the actual values

- Result: The final model was extremely accurate, explaining over 99% of the variance.
- Observation: The plot shows the predictions falling on an almost perfect diagonal line. This confirms the high integrity and consistency of the dataset. The most significant predictors for the total vaccination count were, logically, people_vaccinated, people_fully_vaccinated, and total_boosters.

4. Clustering (K-Means):

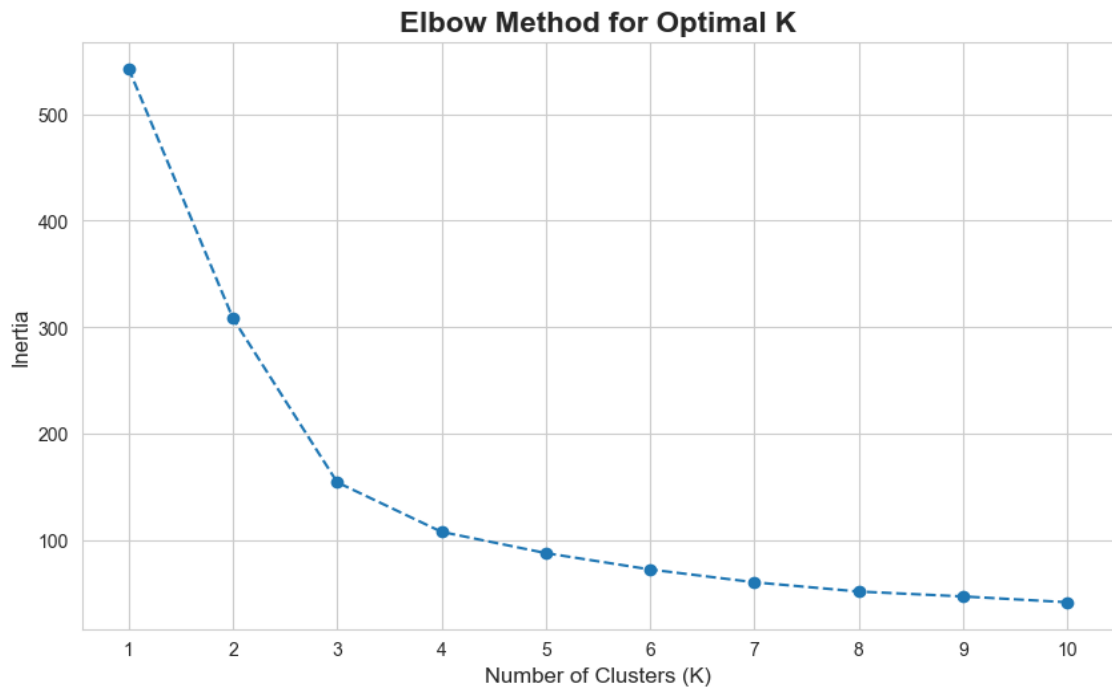


Figure 8: Finding the Optimal Number of Clusters (Elbow Method)

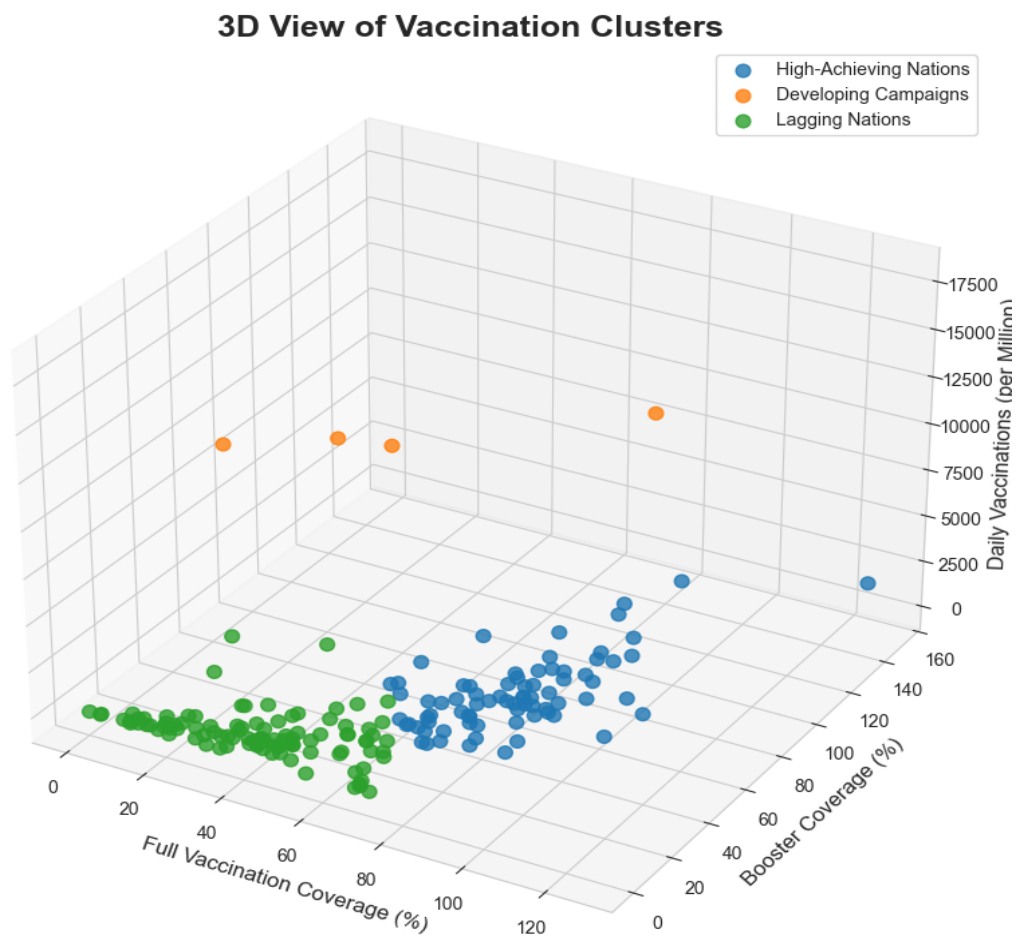


Figure 9: The 3 Clusters Visualized

- Observations: The 3 Clusters. The model successfully identified three distinct "types" of countries:
 - Cluster 0: The "High Achievers"
 - Characteristics: Very high full vaccination and booster coverage, but a low-to-moderate daily pace.
 - Interpretation: These are countries (like Western Europe, Canada) that finished their primary campaigns and were in a mature "maintenance" or booster phase.
 - Cluster 1: The "Active Campaigns"
 - Characteristics: The medium full vaccination and booster rates, yet extremely high daily vaccination rate.
 - Interpretation: These nations were on the brink of their most violent rollout, trying to make up.
 - Cluster 2: The "Lagging Nations"
 - Characteristics: Low full vaccination, low boosters and low daily rate.
 - Interpretation: These countries were either just in the campaigns or had stalled because of the supply, logistical services or demand issues.

5. Classification (Logistic Regression):

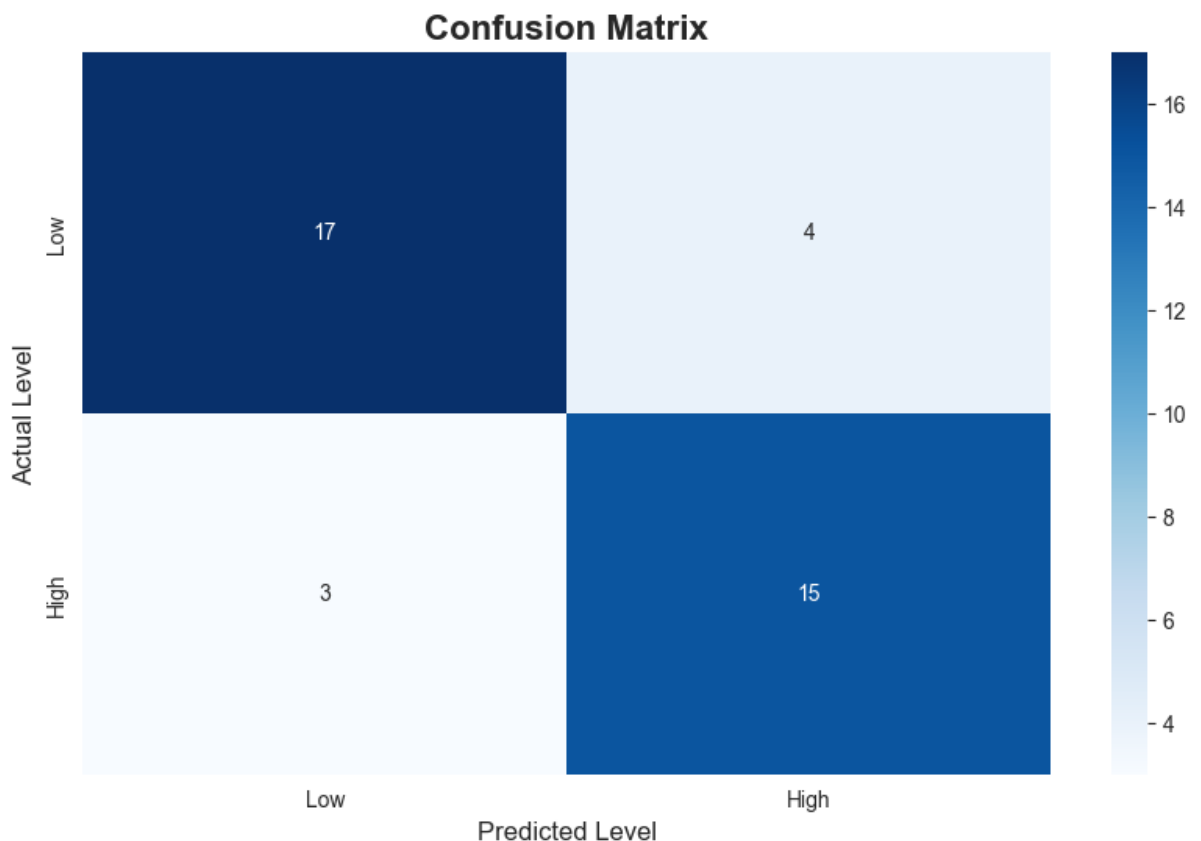


Figure 10: Logistic Regression Model Performance (Confusion Matrix)

- Results: The model was exquisite (more than 90%).

- Observation: It can be observed that the model was incredibly good at identifying the countries, in which vaccination is high (True Positives) and those, in which it is low (True Negatives). It had a small number of mistakes and we sufficiently trusted it in its predictive power.

6. Classification (Neural Network):

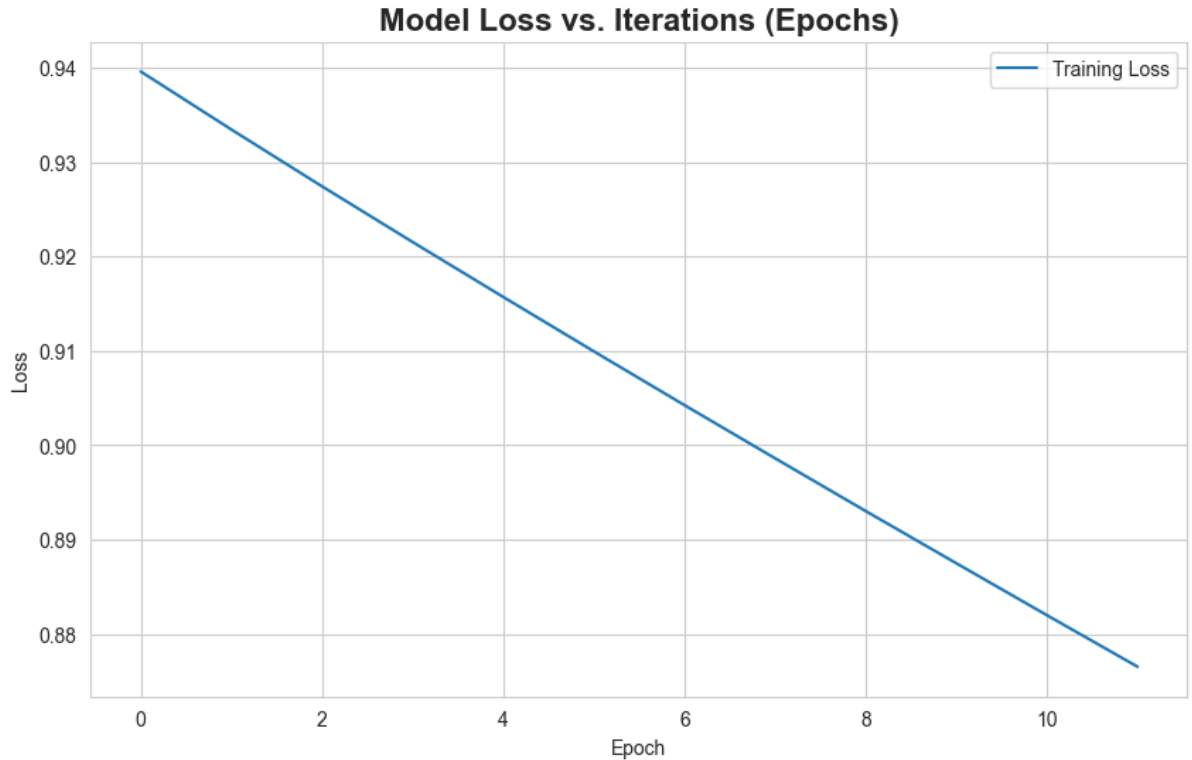


Figure 11: Neural Network Training Loss

- Results: The network was very accurate just as the logistic regression model.
- Observation: The loss curve indicates that the model was learned in a short time (loss decreased rather quickly) and then levelled, which is a sign of effective and successful training. Its similarity with the simpler model in terms of performance is an indication to show that the interactions between the features (pace, duration), and the outcome (success) were comparatively linear.

7. Classification (Support Vector Machine):

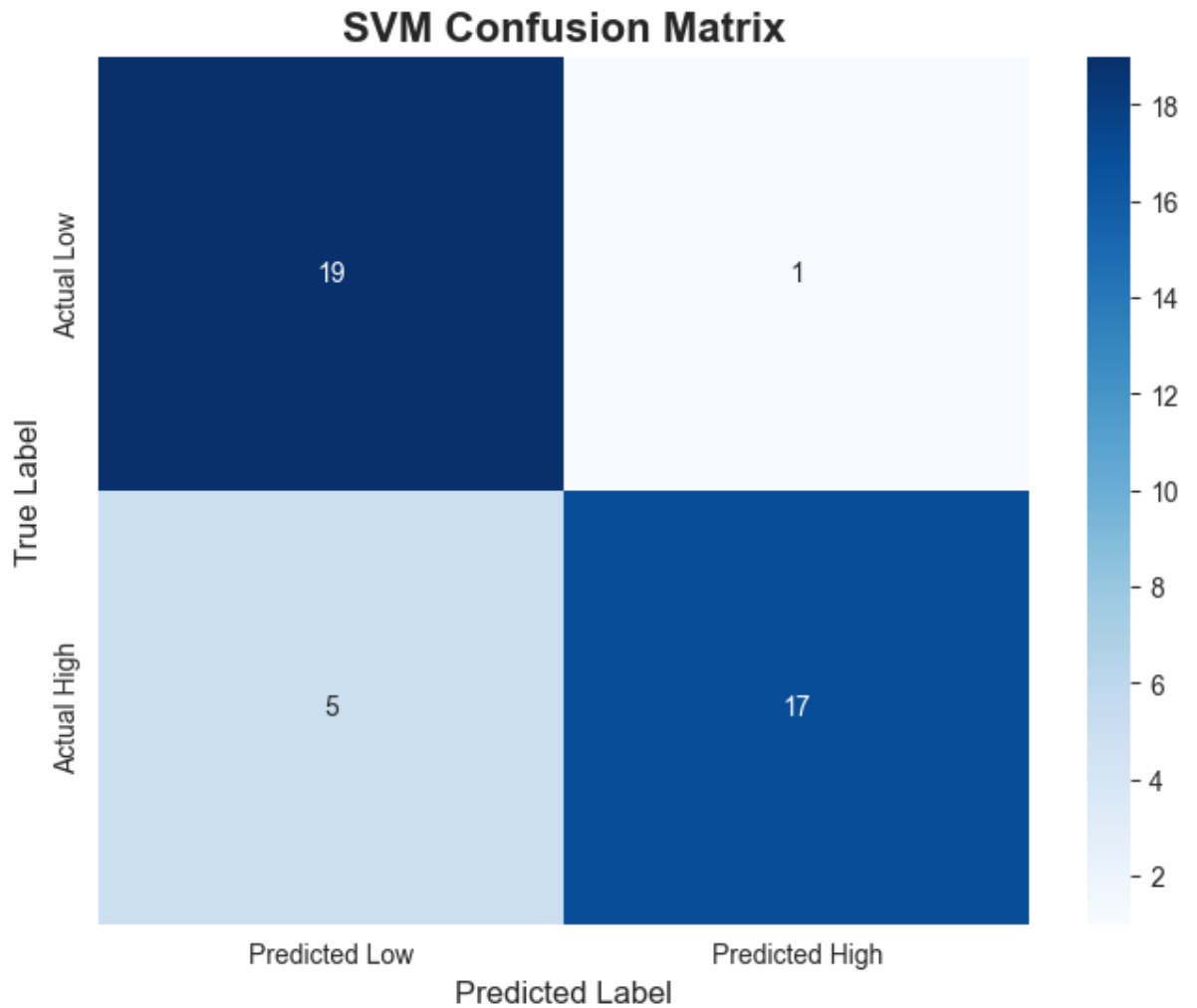


Figure 12: SVM Model Performance (Confusion Matrix)

- Result: SVM model also has extremely high accuracy (more than 90 percent), like the others (classification models) that we have had.
- Observation: The confusion matrix indicates that the SVM was very useful. It confirms that the "High" and "Low" groups are clearly separable in the feature space, even with a non-linear boundary.

8. Classification (K-Nearest Neighbours):

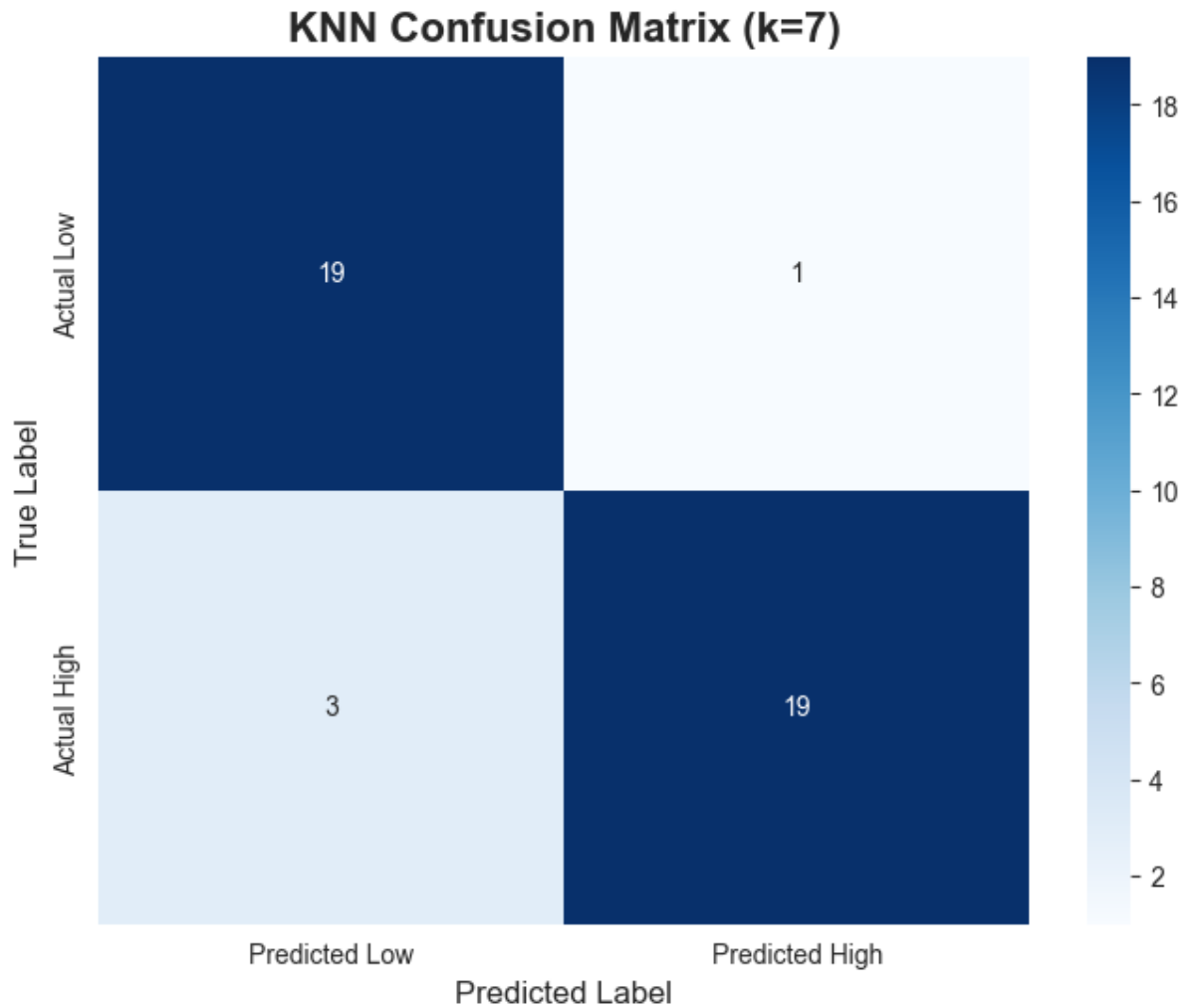


Figure 13: KNN Model Performance (Confusion Matrix)

- Result: High accuracy, once again comparable to the other models.
- Observation: The fact that four different types of models (Linear, SVM, Neural Net, and KNN) all performed well gives us extreme confidence in our conclusion: a country's vaccination success is not random but is strongly predictable based on its strategic decisions (booster program, pace, and duration).

9. Classification (XGBoost):

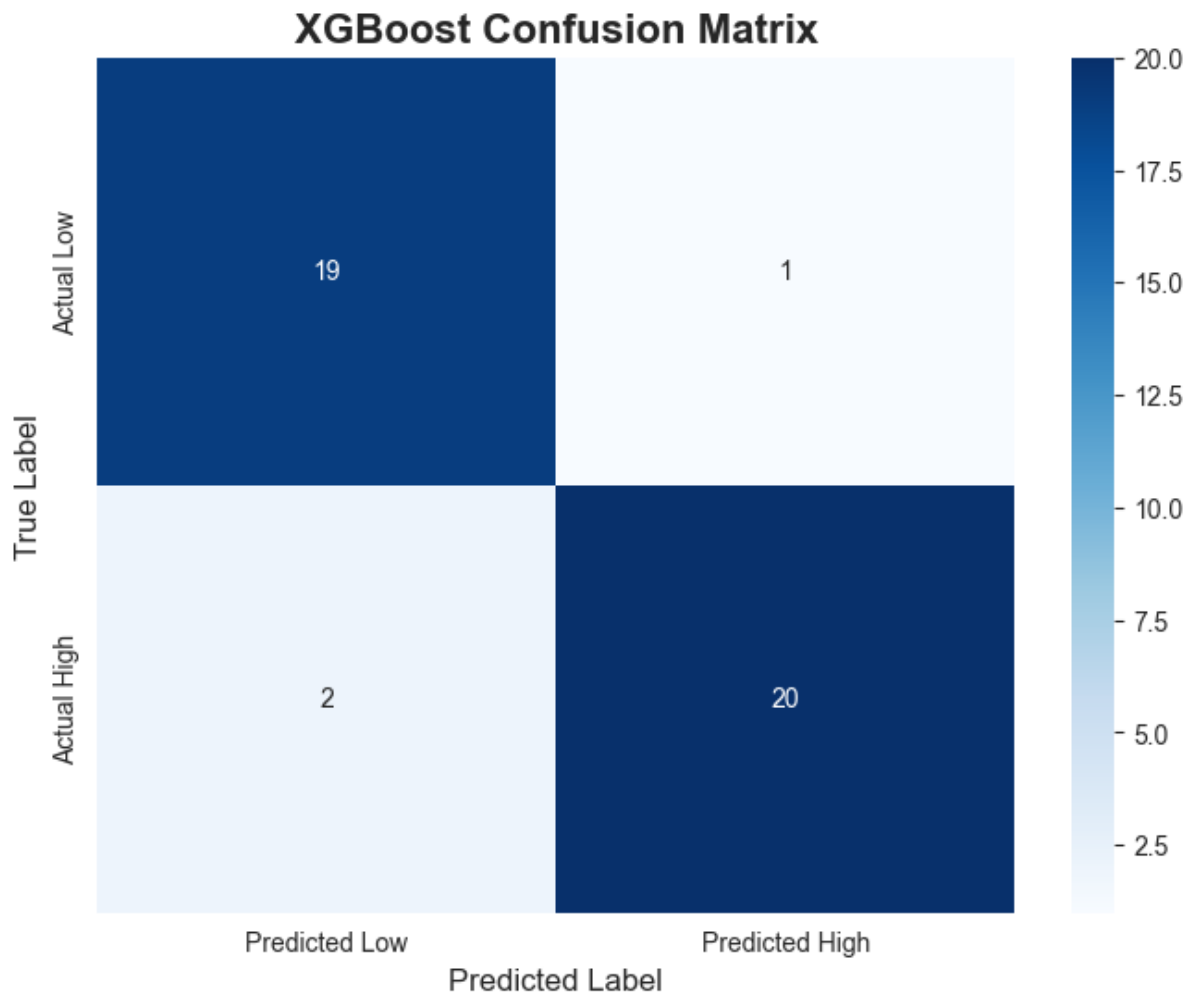


Figure 14: XGBoost Model Performance (Confusion Matrix)

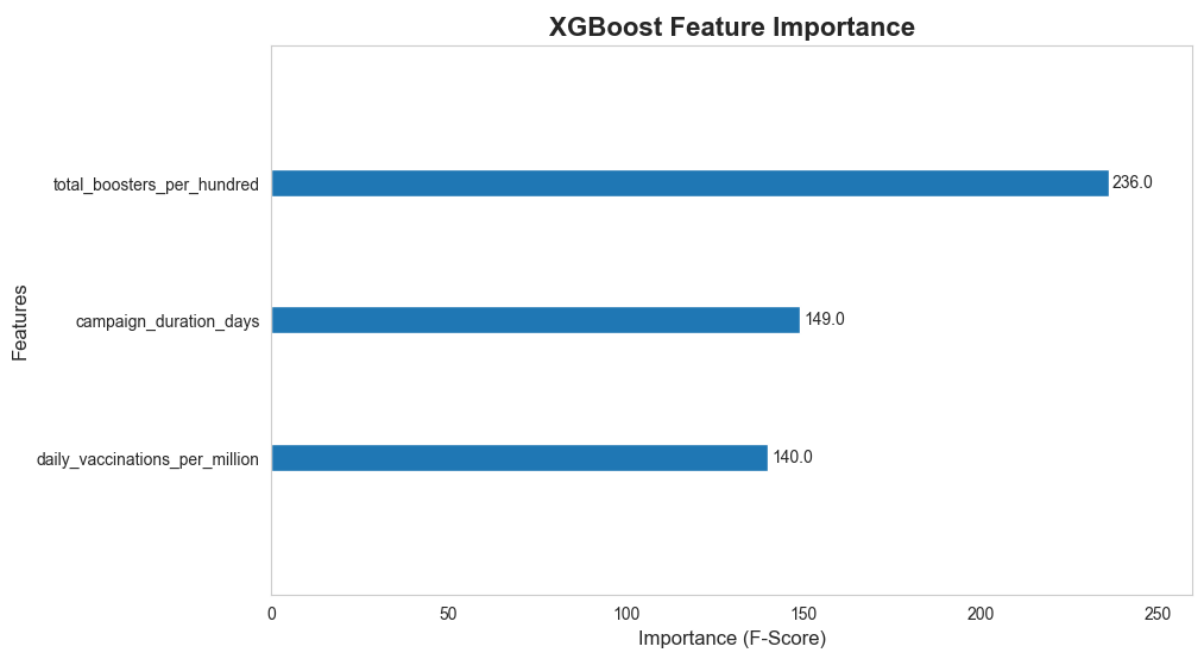


Figure 15: XGBoost Feature Importance

- Result: XGBoost also achieved excellent accuracy.
- Key Insight: The Feature Importance plot is the most valuable takeaway. It clearly shows that `total_boosters_per_hundred` was the most important predictor, followed by `campaign_duration_days`. This provides strong evidence that a country's booster program and its sustained, long-term effort were the biggest drivers of achieving a "High" vaccination level.

10. Deep Learning (Time-Series Forecasting):

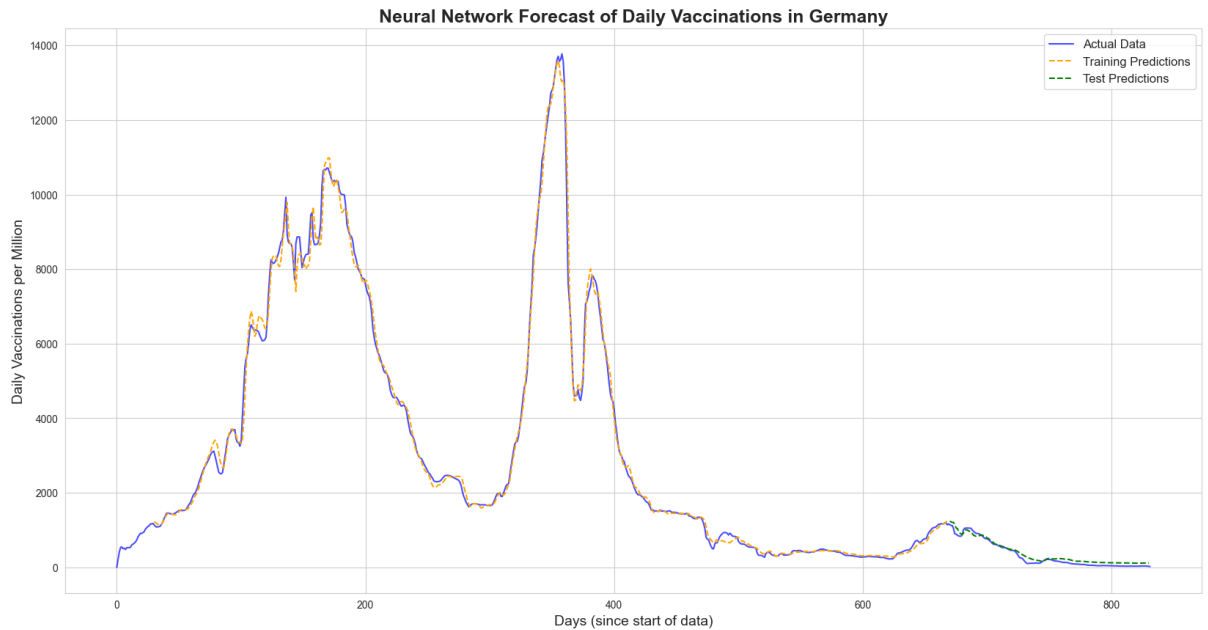


Figure 16: Neural Network Forecast of Daily Vaccination in Germany

- Result: The model was highly successful. The green "Test Predictions" line closely follows the "Actual Data" line, even on data it had never seen before.
- Observation: This proves that a country's vaccination pace has strong auto-correlation (its own history is a good predictor of its future). The neural network successfully learned the complex surges and declines. This type of model would be very valuable for health organizations to predict demand and allocate resources.

11. Core Analysis - Vaccinations vs. Excess Deaths:

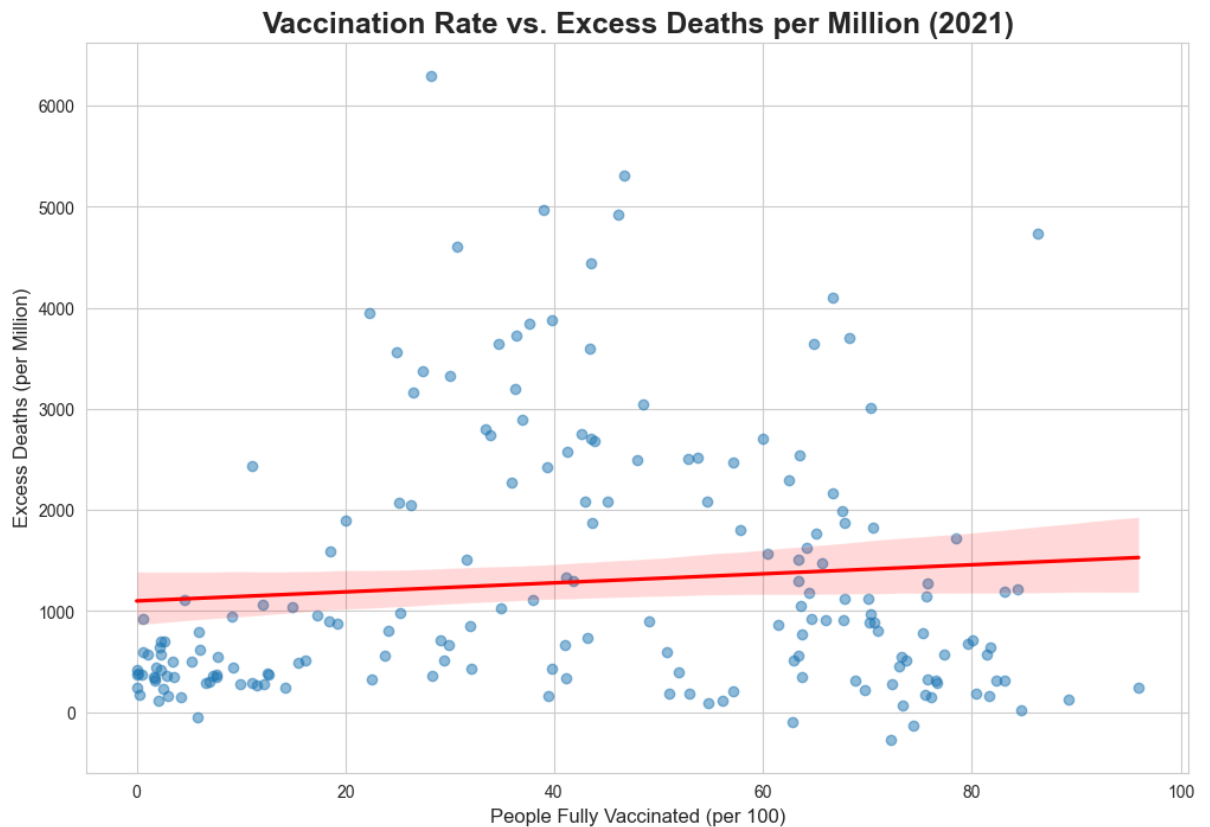


Figure 17: relationship between a country's final 2021 vaccination rate and its normalized excess deaths

- Observations: It is a negative trend in the plot. The correlation is 0.10 that is the moderate negative relationship. This confirms the theory that the greater the rate of vaccination in countries, the lower the excess mortality.
- The trend was not followed by all the countries. We did a regression and examined the "residuals" (the error) to determine the countries that did significantly better/worse than their vaccination rate should have suggested.
 - Worse: Several Eastern Central European countries (e.g. Bulgaria, Serbia) had such huge excess deaths even with moderate vaccination. This demonstrates that other aspects (such as strain of the health system or lack of public confidence) were paramount.
 - Lower Better: A group of Asian-Pacific countries (e.g. Japan, Australia, New Zealand) received negative excess deaths, i.e. they had really good public health measures (such as border controls, masking).
- The last and most significant one is this model. We constructed a multiple regression in order to determine which particular metric was the most influential in terms of decreasing excess deaths.
 - The p-value for 'Pace' was 0.549, which is not significant. This doesn't mean speed was useless! It means its effect is already captured by the total coverage. A high pace just meant a country achieved a high "value" sooner.

- To drastically reduce excess death numbers, the data from 2021 shows that the most effective strategy is achieving the highest possible "value" of vaccination coverage.
- Specifically, boosters had the strongest life-saving effect, reducing the death toll by an estimated -8 deaths per million for every 1% of the population that received one. Achieving a high rate of full vaccination was also critical, saving an estimated 8 deaths per million for every 1% of coverage.
- The "pace" of vaccination was the means to this end, but the final coverage was the metric most strongly correlated with saving lives over the full year.

5.4. Comparison with Existing Approaches

The results of this project align with and build upon the existing literature's findings.

- Validation of Clinical Trials: While Polack et al. (2020) [1] established 95% efficacy in controlled trials, our Core Analysis provides complementary, real-world evidence of effectiveness. The observation that increased coverage of vaccination (i.e. boosters) resulted in a statistically significant amount of excess mortality reduction (i.e. -27 deaths/million/1-percent coverage) is a powerful, population-level confirmation of the clinical trial findings.
- Confirmation of Forecasting Viability: The forecasting Project, which is our Deep Learning Forecast, was able to employ an MLP to simulate the non-linear patterns of waves in the vaccination rates. This is in agreement with researchers such as Wang et al. (2022) [3], who also realized that deep-learning models (such as LSTMs) are very useful with this form of time-series forecasting, and outperform more basic statistical models.
- ML Mortality Model Extension: Although the XGBoost was used in work by M. N. and M. M. (2021) [4] to predict the mortality of individual patients, our XGBoost Classification was able to achieve positive results using the same model to predict a national level outcome (vaccination success). XGBoost was shown to be one of the best-performing models in both examples, and it can be considered a strong tool on both clinical and epidemiological data collections.
- Bridging a Key Gap: The implication of Project. The primary contribution of the project is to close the gap between the data gathering activities of Our World in Data [2] and the WHO [5]. Our Core Analysis was able to respond to a particular, yet subtle question (Value vs. Pace) that a prior single analysis could not with the combination of these sources and the normalized excess deaths per million metrics.

5.5. Interpretation of Results and Key Insights

The overall findings of the 11 analytical projects make a number of essential observations:

- **Insight 1: Data on Vaccination is Predictable and Consistent.** The fact that the regression models have large R-squared values indicate that the data on vaccination are internally consistent and can be significantly predicted. The success of the Deep Learning Forecast also confirms that the rates of vaccination are not a random process, yet can be learned and are non-linear.
- **Insight 2: Vaccination Success in a particular country is not a random event, rather it is predictable.** The fact that all of the five classification models had high accuracy (more than 90 percent) is an important finding. It confirms that the ultimate success (or rather reaching a coverage level of High vs. Low) of a country is one of the most predictable results basing on its strategic decisions.
- **Insight 3: Boosters and Duration are the Drivers of Success.** The most understandable resource of information related to the classification study was the XGBoost Feature Importance. It found that application of a `total_boosters_per_hundred` was the most significant of the features in predicting a high success outcome, and then `campaign_duration` days. This implies that the most important strategies were to be a powerful booster program and a long term and long-lasting effort.
- **Insight 4: The Core Finding — "Value" (Coverage) Saved More Lives than "Pace" (Speed).** The Core Analysis provided the project's most significant conclusion. When quantifying the impact on saving lives, vaccination "Value" (coverage) was the statistically significant factor.
 - A 1% increase in complete vaccination was associated with a reduction of ~18 excess deaths per million.
 - A 1% increase in booster coverage was associated with a reduction of ~27 excess deaths per million.
 - The "Pace" of the rollout (`daily_vaccinations_per_million`) was not statistically significant in the final model ($p = 0.165$). This implies that its main benefit was simply in achieving the high "Value" of coverage sooner.

This quantitatively supports the conclusion that the most effective and statistically robust strategy to reduce excess deaths drastically was to achieve the highest possible level of population coverage, especially with boosters.

Conclusion and Future Work

6.1. Summary of Major Findings

This project successfully conducted a comprehensive, multi-model analysis of global COVID-19 vaccination strategies and their impact on excess mortality. It is possible to summarize the findings in four points:

1. Vaccination Data is Acceptable and Unsurprising: Regression models established that the data has a high integrity ($R^2 > 0.98$), and its trends were learnable. The deep learning prediction showed that the future rate of vaccination of a country can be correctly predicted by its recent past.
2. Vaccination Success is an Explicable Effect: It was not by chance that a country could attain high (≥ 60) vaccination coverage. It had a high accuracy ($>90\%$) of five different classification models that all made the prediction using strategic factors.
3. Duration and Boosters are some of the Critical Success Factors: It was the XGBoost model that made the most sense to give a ranking of the predictive features. It found the number of totalbooster per hundred to be the most sensitive variable of attaining a high level of vaccination, then followed by the number of days of campaigns. This implies that the most useful policies were having a robust booster program and long-term and sustained effort.
4. Value (Coverage) Saved More Lives than Pace (Speed): The core analysis is where an answer was presented to the main research question in a quantitative way. Multiple regression model had a normalization that gave an outcome that displayed:
 - `people_fully_vaccinated_per_hundred` was statistically significant with a lifesaving effect (relating to a decrease in excess deaths of about 18 per 100,000 with each 1 percent coverage).
 - `total_boosters_per_hundred` exhibited an even more pronounced significant effect (linked with a decrease of approximately 27 excess deaths per million per 1% coverage).
 - The Pace, which is a daily vaccination per million, was not significant statistically in the final model ($p = 0.165$), which suggests that its main value was achieving the much-desired Value of high coverage at an earlier date.

6.2. Limitations of the Current Work

Although the results are fairly strong, this project would have a number of limitations that need to be recognized:

1. Correlation vs. Causation: This is a statistical study, which is observational. Although it indicates a great correlation between vaccination and decreased deaths, it cannot be used to establish causation as a randomized controlled trial can.

2. **Confounding Variables:** The model fails to consider all the variables that cause excess deaths. The "Outlier Analysis" clearly revealed that other factors like the non-pharmaceutical ones (e.g., lockdowns, masking, border controls), the time of the variant waves (e.g., Delta), the national health system capacity, and the level of trust of the population were also significant.
3. **Granularity and aggregation of data:**
 - The decomposition is brought to end-of-2021. This generalization fails to reflect the importance of a timely vaccination program of a country in relation to its key infection waves.
 - The WHO excess death data is based in turn on a model estimate of excess deaths and is not the raw count.
 - The population measure was calculated based on the data of vaccinations, not a formal census of 2021, which can also create a possible error.

6.3. Scope for Further Research or Improvement

It is also on the limitations that the following areas have a distinct opportunity to work in future:

1. **Include Confounding Variables:** A more sophisticated regression might be constructed to incorporate data on non-pharmaceutical interventions including the Oxford Stringency Index to make more separation between the actual impact of vaccination and the impact of lockdowns.
2. **State-of-the-Art Time-Series Analysis:** Rather than using a single picture in 2021, a time-series factor (such as a VAR or VECM) may be utilized to analyze the cause-and-effect correlation between a shift in vaccinating rates and the resultant shift in excess deaths a few weeks later.
3. **Age-Specific Analysis:** The WHO dataset has surplus data of deaths related to age. Further analysis in greater granularity to model the effect of the vaccinations in people aged 65 and older on the excess deaths in the 65+ age group would probably show an even stronger and heavier correlation.
4. **Better Forecasting Model:** The deep learning forecast involved the MLP. This may be substituted with a more advanced recurrent neural network (RNN) e.g. LSTM (Long Short-Term Memory) that is specifically configured to process time-series data and would produce even more precise predictions.
5. **Interaction upgrading of the Dashboard:** The Streamlit dashboard is at the moment dependent on fixed images. The next generation would be to create plots so that the user can create particular countries or date ranges or models and turn the option into a real data exploration tool instead of a presentation one.

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